

Junhwan Heo

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Education

Korea University, Bachelor of Science in Data Science Mar 2021 – Present

• Relevant Coursework:

- **AI:** Artificial Intelligence, Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, Introduction to Computer Vision and Its Applications.
- **Mathematics & Statistics:** Calculus and Practice I, Mathematical Statistics, Regression Analysis, Multivariate Statistical Analysis, Statistical Data Science I, Statistical Computing Methods.

Gwangju Science Academy for the Gifted, High School Diploma Mar 2018 – Feb 2021

- Specialized secondary education program for gifted students in science and mathematics.

Research Experience

Undergraduate Research Intern, Universal Transfer Learning Lab, Korea University June 2025 – Present

Advisor: Professor Donghyun Kim

- Conducting research on **scene graph generation (SGG)**
- Investigating the effect of **synthetic data augmentation** on relational reasoning and generalization in SGG.
- Exploring **diffusion-based generation of relational datasets** as a potential extension for improved visual understanding.

Projects

Lookback Lens and LAG Integration 2025

- Designed **RELF (Retrieval-Enhanced Lookback Fusion)** decoding framework combining retrieval-augmented generation (RAG) with Lookback Lens for factuality control in small language models.
- Conducted extensive experiments on **Natural Questions Open** dataset with *LLaMA-2-7B* model to evaluate hallucination and factual accuracy.
- Tools Used: Python, Hugging Face Transformers, FAISS, LangChain

Low-light Image Recognition 2025

- Conducted a project on improving image classification performance under low-light conditions.
- Proposed a **dual-channel architecture** combining raw RGB inputs with edge-enhanced representations (Canny maps) to increase robustness to illumination variation.
- Tools Used: Python, TensorFlow, OpenCV

Lecture Review Clustering 2022

- Crawled university lecture reviews from the **KLUE platform** using *BeautifulSoup (bs4)* for data collection and cleaning.
- Performed **KMeans clustering** to group similar lecture reviews based on content similarity and sentiment patterns.
- Tools Used: Python, scikit-learn, BeautifulSoup

Public Mask Inventory Mobile App 2020

- Developed a Kotlin-based Android application that visualizes nearby mask stock levels using public APIs.
- Tools Used: Kotlin, REST API

Deep Learning-based Missing Value Imputation for Air Quality Data 2019

- Conducted an R&E research project on imputing missing values in air quality and meteorological datasets to improve the reliability of environmental monitoring.
- Tools Used: Python, PyTorch, Scikit-learn

Other Experience

Military Service, Republic of Korea Air Force, 17th Fighter Wing
Information Systems Specialist

Apr 2023 – Jan 2025

- Managed and maintained the Air Force internal network and intranet systems (AFCCS).
- Operated and monitored mission-critical servers and communication infrastructure.
- Collaborated with IT officers to ensure cybersecurity compliance and system stability.

Technologies

Languages: C++, C, SQL, R, Python, Kotlin

ML/DL: Pytorch, Tensorflow